

Conservation and Convertibility as Structural Necessities of Persistent Systems

From Noether's Theorem to a General Theory of
System Persistence Across Physical, Biological, Cognitive, and Social
Domains

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Abstract

Conservation laws and mass–energy equivalence are foundational results in physics. This paper argues that they are instances of deeper *structural necessities* governing any complex system that persists through time in a changing environment. Two meta-principles are identified and formally derived from a minimal persistence axiom: (i) **Conservation Necessity**—that every persistent system must track and balance a set of identity-constitutive quantities, and (ii) **the Convertibility Principle**—that structure and process must be interconvertible, with a domain-specific conversion factor γ that bounds the system's resistance to transformation. The paper introduces a state-space formalism accessible to researchers across disciplines and proves that systems possessing conservation without convertibility become rigid and brittle, while those possessing convertibility without conservation become dissipative and identity-less; only systems satisfying both occupy a viable persistence region. These results are instantiated across physics, biology, cognitive science, economics, and social systems, with formal structural correspondences established rather than merely asserted by analogy. A comprehensive literature review connects the framework to Noether's theorem, autopoiesis, the free energy principle, ecological rationality, dissipative structures, and general systems theory, while identifying what the present contribution adds to each. The framework is shown to

be self-applicable: its own epistemic limitations are predicted by the constraints it identifies.

Keywords: conservation laws, persistence, complex systems, structure–process duality, viability, self-organization, meta-principles, dynamical systems, cross-domain isomorphism

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1 Introduction

Few results in physics carry the intuitive weight of conservation laws. That energy can be neither created nor destroyed, that momentum persists through every collision, that charge is invariant across all electromagnetic interactions—these are among the first principles taught and the last abandoned. Emmy Noether’s celebrated theorem (Noether, 1918) revealed a deep unity beneath these separate laws: every continuous symmetry of a physical system’s action yields a conserved quantity. Time-translation symmetry yields energy conservation; spatial-translation symmetry yields momentum conservation; gauge symmetry yields charge conservation.

Separately, Einstein’s mass–energy equivalence $E = mc^2$ (Einstein, 1905) demonstrated that what had been regarded as two fundamentally different physical substances—matter and energy—are interconvertible expressions of a single underlying reality. Mass is concentrated energy; energy is liberated mass.

These results are typically understood as properties of the physical world. The present paper advances a different thesis: *conservation and interconvertibility are not merely features of physics but structural necessities of any complex system that persists through time in a changing environment*. Noether’s theorem and mass–energy equivalence are the mathematically precise instances of more general principles that apply wherever persistent organized complexity is found—in cells, brains, institutions, economies, and information systems.

This claim requires care. Cross-domain analogies between physics and social systems have a long and often disreputable history, from social Darwinism to crude thermodynamic metaphors in economics. The present work therefore insists on a higher evidentiary standard: the cross-domain correspondences identified here are *derived* from shared structural constraints, not merely *asserted* by surface analogy. The methodology follows a constraint-derivation approach in which the features of persistent systems are deduced from conditions that *any* system of a given class must satisfy—an approach with precedent in Ashby’s requisite variety (Ashby, 1956), Rosen’s relational biology (Rosen, 1991), and more recently in the free energy principle (Friston, 2010).

The paper is organized as follows. Section 2 reviews the existing literature on conservation, persistence, and cross-domain structural theory. Section 3 introduces the formal framework. Sections 4 and 5 derive the two meta-principles. Section 6 proves that their conjunction defines a viability region. Section 7 instantiates the framework across five domains. Section 8 investigates the domain-specific conversion factor. Section 9 applies the theory to itself. Section 10 situates the results within a broader research program. Section 11 discusses implications and limitations.

2 Literature Review

The aspiration to identify unifying principles across complex systems has a rich intellectual history. This section surveys the relevant traditions, identifies what each contributes, and locates the gap the present paper addresses.

2.1 Conservation Laws in Physics

Noether’s theorem (Noether, 1918) established the foundational link between symmetry and conservation in Lagrangian mechanics. Subsequent developments extended the principle to quantum field theory, where gauge symmetries yield conserved currents via the Ward–Takahashi identities (Ward, 1950; Takahashi, 1957). The deep lesson of this tradition is that conservation is not imposed on physical systems from outside but *follows from the structural properties of the system’s dynamics*—specifically, from its symmetries.

Einstein’s special relativity (Einstein, 1905) unified mass and energy, revealing that apparent ontological categories (“matter” and “radiation”) are interconvertible manifestations of a single quantity. This insight was generalized in general relativity, where the stress–energy tensor $T_{\mu\nu}$ encodes all forms of energy, momentum, and stress in a single geometric object (Misner et al., 1973).

2.2 General Systems Theory and Cybernetics

Von Bertalanffy’s general systems theory (von Bertalanffy, 1968) was among the earliest systematic attempts to identify structural isomorphisms across domains. His central insight—that “there exist models, principles, and laws that apply to generalized systems or their subclasses, irrespective of their particular kind”—directly anticipates the present work, though Bertalanffy’s framework remained largely descriptive rather than deductive.

Ashby’s cybernetics (Ashby, 1956) introduced the law of requisite variety, which can be read as a proto-conservation principle: a controller must possess at least as much variety as the disturbances it regulates. Wiener’s foundational work on feedback systems (Wiener, 1948) established that self-regulating systems necessarily maintain certain internal quantities within bounded ranges—a functional analogue of conservation.

2.3 Autopoiesis and Organizational Closure

Maturana and Varela’s theory of autopoiesis (Maturana & Varela, 1980) formalized the concept of organizational closure: a system that produces and maintains the components that constitute it. This is conceptually close to the conservation principle developed here, as autopoietic systems must conserve their own organization through continuous material

turnover. Varela’s later work on operational closure in the immune and nervous systems (Varela, 1979) extended the concept beyond cellular biology.

Rosen’s work on (M, R) -systems (Rosen, 1991) provided a category-theoretic formalization of organizational closure, demonstrating that the essential property of living systems is not their material composition but their *relational structure*—specifically, the closure of their entailment relations. This anticipates the present paper’s emphasis on structural rather than substrate-specific characterizations.

2.4 Dissipative Structures and Self-Organization

Prigogine’s theory of dissipative structures (Prigogine & Stengers, 1984) demonstrated that open systems far from thermodynamic equilibrium can spontaneously generate and maintain organized patterns by continuously dissipating energy. This introduced a crucial nuance: in open systems, conservation does not mean stasis but rather *dynamic balance*—a continuous throughput of matter and energy that maintains structural integrity. Haken’s synergetics (Haken, 1983) provided a complementary mathematical framework for understanding how order emerges at macroscopic scales through the enslaving of fast variables by slow order parameters.

Kauffman’s work on self-organization at the edge of chaos (Kauffman, 1993) suggested that complex adaptive systems naturally evolve toward a regime that is neither too ordered nor too disordered. This “edge of chaos” hypothesis resonates with the present paper’s claim that viable systems occupy a bounded region in the conservation–convertibility space.

2.5 The Free Energy Principle

Friston’s free energy principle (Friston, 2010, 2013) provides the closest existing framework to the one developed here. It proposes that any self-organizing system that is at equilibrium with its environment must minimize a free energy functional—or equivalently, that such systems resist the natural tendency toward disorder by maintaining a generative model of their environment. The principle has been applied to neural processing (Friston, 2010), cellular biology (Friston, 2013b), and social systems (Ramstead et al., 2018).

The free energy principle can be understood as a specific formalization of the conservation necessity: the quantity being conserved is the system’s structural integrity (formalized as surprise or negative log-evidence), and the mechanism of conservation is active inference. The present paper generalizes this by abstracting away from the specific free-energy formalism to identify the bare structural requirements—conservation and convertibility—that any persistent system must satisfy, regardless of whether its dynamics can be expressed as free-energy minimization.

2.6 Ecological Rationality and Adaptive Heuristics

Gigerenzer and colleagues’ ecological rationality program (Gigerenzer et al., 1999; Gigerenzer & Gaissmaier, 2011) demonstrated that cognitive heuristics and biases, traditionally viewed as failures of rational inference, are often well-adapted solutions to decision problems in resource-constrained environments. Todd and Gigerenzer’s “fast and frugal heuristics” (Todd & Gigerenzer, 2012) showed that simple decision rules can match or exceed the performance of complex optimization algorithms when ecological structure is taken into account.

This tradition supports the present paper’s treatment of cognitive biases as instances of the structure–process duality: mental models (structure) are efficient compressions of past inferential activity (process), and their apparent distortions reflect optimal trade-offs under constraint rather than irrational errors.

2.7 Process Philosophy and Organizational Theory

Whitehead’s process philosophy (Whitehead, 1929) argued that reality is fundamentally composed of processes rather than substances—that what we perceive as enduring objects are in fact patterns of becoming. This ontological claim finds formal expression in the present paper’s convertibility principle, where structure is characterized as temporarily stabilized process.

In organizational theory, Weick’s concept of “organizing” versus “organization” (Weick, 1979) captures a similar insight: institutions are not static entities but ongoing achievements of coordinated action. March’s exploration of the tension between exploitation (leveraging existing structures) and exploration (generating new processes) (March, 1991) maps directly onto the conservation–convertibility duality.

2.8 Category Theory and Structural Mathematics

Recent work in applied category theory has provided formal tools for establishing structural correspondences across domains. Baez and Stay’s “Rosetta Stone” (Baez & Stay, 2011) showed that physics, topology, logic, and computation share common categorical structures. Spivak’s operads framework (Spivak, 2014) provides a mathematical language for composing complex systems from modular components. These approaches demonstrate that cross-domain structural isomorphisms can be made rigorous rather than merely metaphorical.

2.9 Gap in the Literature

Despite the richness of these traditions, no existing framework simultaneously: (a) derives conservation and convertibility as *joint* structural necessities from a single persistence

axiom; (b) provides a formal characterization of the viability region defined by their intersection; (c) introduces and analyzes the domain-specific conversion factor γ between structure and process; and (d) demonstrates the self-applicability of the resulting theory. The present paper addresses this gap.

3 Formal Framework

This section introduces the mathematical apparatus used throughout the paper. The definitions are drawn from dynamical systems theory but are presented with the aim of making the formal content accessible to researchers whose primary training is in the biological, cognitive, or social sciences.

3.1 Systems, States, and Trajectories

We begin with the concept of a system occupying a state that changes over time.

Definition 3.1 (System). A **system** is a triple $\mathcal{S} = (X, T, \Phi)$ where:

- X is a **state space**—the set of all configurations the system can occupy.
- T is a **time set**, typically $\mathbb{R}_{\geq 0}$ (continuous time) or $\mathbb{N}_0$ (discrete time).
- $\Phi : T \times X \rightarrow X$ is a **dynamics map** satisfying $\Phi(0, x) = x$ and $\Phi(t_1, \Phi(t_2, x)) = \Phi(t_1 + t_2, x)$ for all $t_1, t_2 \in T$ and $x \in X$.

The state space X can be thought of as a landscape in which every possible configuration of the system—every arrangement of its components, every distribution of its resources—corresponds to a single point. The dynamics map Φ describes how the system moves through this landscape: given a starting configuration x and an elapsed time t , the expression $\Phi(t, x)$ specifies where the system will be. The two conditions on Φ ensure, respectively, that the system starts where it starts and that the system’s future depends on its present state rather than on how it arrived there.

Definition 3.2 (Trajectory). Given an initial state $x_0 \in X$, the **trajectory** of the system is the function $\tau_{x_0} : T \rightarrow X$ defined by $\tau_{x_0}(t) = \Phi(t, x_0)$.

A trajectory is simply the system’s history—the complete sequence of states it passes through starting from a given initial configuration.

3.2 Observable Quantities

Not every aspect of a system’s state is equally important for understanding its behavior. We formalize the properties we track as observable quantities.

Definition 3.3 (Observable). An **observable** on a system $\mathcal{S} = (X, T, \Phi)$ is a measurable function $f : X \rightarrow \mathbb{R}$.

An observable extracts a single number from a system’s configuration. In a biological context, an observable might measure body temperature, blood glucose concentration, or population size. In an economic system, it might measure total capital, liquidity, or the Gini coefficient. In a cognitive system, it might measure prediction error or attentional capacity.

Definition 3.4 (Conserved Quantity). An observable $f : X \rightarrow \mathbb{R}$ is **conserved** under Φ if for all $x \in X$ and all $t \in T$,

$$f(\Phi(t, x)) = f(x). \quad (1)$$

A conserved quantity is one whose value does not change as the system evolves. In physics, total energy in a closed system is conserved. This notion will be generalized below to accommodate open systems where exact conservation is replaced by *bounded* conservation—a quantity that remains within a viable range rather than at a fixed value.

3.3 Persistence and the Viability Set

The central concept of this paper is persistence: a system’s capacity to maintain its identity over time.

Definition 3.5 (Identity-Constitutive Set). Given a system $\mathcal{S} = (X, T, \Phi)$, an **identity-constitutive set** is a collection of observables $\mathcal{F} = \{f_1, \dots, f_n\}$ such that the system retains its identity if and only if all f_i remain within specified bounds. The **viability set** is:

$$\mathcal{V} = \{x \in X \mid \alpha_i \leq f_i(x) \leq \beta_i \text{ for all } i = 1, \dots, n\}. \quad (2)$$

The viability set $\mathcal{V}$ represents the region of state space within which the system remains itself. A human organism persists as long as body temperature, blood pH, oxygen saturation, and other vital parameters remain within their respective bounds. An institution persists as long as its constitutive norms, roles, and resource flows remain functional. The bounds α_i and β_i are not arbitrary but are determined by the system’s structure and the constraints of its environment.

Definition 3.6 (Persistence). A system $\mathcal{S}$ **persists** over a time interval $[0, T^*]$ from initial state x_0 if:

$$\Phi(t, x_0) \in \mathcal{V} \quad \text{for all } t \in [0, T^*]. \quad (3)$$

A system persists **indefinitely** if (3) holds for all $t \in T$.

Persistence, then, is the system’s trajectory remaining within the viability set. This is the formal expression of the intuitive notion that a system “continues to exist”—not in the trivial sense that its atoms persist, but in the meaningful sense that its characteristic organization is maintained.

3.4 Open Systems and Environmental Coupling

Real systems interact with their environments. We model this as follows.

Definition 3.7 (Open System). An **open system** is a tuple $\mathcal{S}_{\text{open}} = (X, E, T, \Phi)$ where E is an **environment space** and $\Phi : T \times X \times E \rightarrow X$ is a dynamics map parameterized by environmental states. The environment evolves according to its own dynamics $\Psi : T \times E \rightarrow E$, and we write $e(t) = \Psi(t, e_0)$ for the environmental trajectory.

The distinction between closed and open systems is critical. In a closed system, exact conservation is possible: a quantity can maintain precisely the same value forever. In an open system—and every real biological, cognitive, and social system is open—exact conservation is the exception. What persists instead is *regulated conservation*: the system actively works to keep its identity-constitutive quantities within viable bounds despite environmental perturbations.

3.5 Structure and Process

The second meta-principle requires a formal distinction between static and dynamic aspects of a system.

Definition 3.8 (Structural and Processual Decomposition). Given a system $\mathcal{S} = (X, T, \Phi)$, a **structural-processual decomposition** is a pair of subspaces (S, P) of X such that:

1. X can be expressed (at least locally) as a product: $X \cong S \times P$, where S encodes the system’s **structural** (slowly changing, high-inertia) components and P encodes its **processual** (rapidly changing, low-inertia) components.
2. The decomposition is **timescale-relative**: what counts as structure at one temporal resolution may be process at another.

Consider a biological organism. At the timescale of seconds, the skeleton is structure and neural firing patterns are process. At the timescale of decades, the skeleton itself is process (it is continuously rebuilt through osteoclast and osteoblast activity), and what persists as structure is the genetic program. This timescale-relativity is not a weakness of the decomposition but a feature predicted by the theory: the boundary between structure and process is itself a system property, not an observer convention.

4 The Conservation Necessity

We now derive the first meta-principle: that any persistent system must maintain conservation-like constraints on its identity-constitutive quantities.

Axiom 4.1 (Persistence Axiom). Let $\mathcal{S}_{\text{open}} = (X, E, T, \Phi)$ be an open system with viability set $\mathcal{V} \subset X$. The system is **observed to persist** over the interval $[0, T^*]$.

This axiom makes no claim about *why* the system persists—it is a selection statement. Among all possible systems and initial conditions, we consider only those that have in fact persisted. This is the formal expression of an observational fact: if we are studying a system (a cell, a brain, a society, an ecosystem), it has, by definition, persisted long enough to be studied.

Theorem 4.1 (Conservation Necessity). *Let $\mathcal{S}_{open}$ be an open system satisfying Axiom 4.1 with identity-constitutive observables $\mathcal{F} = \{f_1, \dots, f_n\}$ and viability set $\mathcal{V}$ as in Definition 3.5. Then for each $f_i \in \mathcal{F}$ and for all $t \in [0, T^*]$:*

$$\alpha_i \leq f_i(\Phi(t, x_0, e(t))) \leq \beta_i. \quad (4)$$

Moreover, defining the **conservation functional** $C_i : [0, T^*] \rightarrow \mathbb{R}$ by

$$C_i(t) = f_i(\Phi(t, x_0, e(t))) - f_i(x_0), \quad (5)$$

we have $|C_i(t)| \leq \beta_i - \alpha_i$ for all t . The trajectory of each identity-constitutive observable is bounded.

Proof. By Axiom 4.1, $\Phi(t, x_0, e(t)) \in \mathcal{V}$ for all $t \in [0, T^*]$. By Definition 3.5, membership in $\mathcal{V}$ requires $\alpha_i \leq f_i(x) \leq \beta_i$ for all i . Thus (4) follows directly.

For the bound on C_i : since $f_i(\Phi(t, x_0, e(t))) \in [\alpha_i, \beta_i]$ and $f_i(x_0) \in [\alpha_i, \beta_i]$ (as $x_0 \in \mathcal{V}$), the maximum deviation is $|C_i(t)| \leq \beta_i - \alpha_i$. $\square$

Theorem 4.1 may appear tautological—and in a narrow sense it is, since it follows directly from the definitions. Its significance lies not in logical surprise but in *reframing*: it shows that what physics calls “conservation” (the invariance of certain quantities) is a special case of what persistence *requires* of any viable system. In a closed physical system, the bounds $[\alpha_i, \beta_i]$ collapse to a point: $\alpha_i = \beta_i = f_i(x_0)$, recovering exact conservation. In open biological, cognitive, and social systems, the bounds define a range—and the system must actively regulate its observables to remain within that range.

The critical implication is that *any persistent system must perform a form of accounting*. The specific quantities that are tracked vary across domains—energy and momentum in physics, metabolite concentrations in biology, prediction error in cognition, trust and legitimacy in institutions—but the structural requirement is universal.

Corollary 4.1 (Active Regulation in Open Systems). *If the environment $e(t)$ includes perturbations that would, in the absence of system response, drive some f_i outside $[\alpha_i, \beta_i]$, then persistence requires the existence of a regulatory mechanism: a component of Φ that counteracts environmental perturbations to maintain $f_i \in [\alpha_i, \beta_i]$.*

This corollary connects conservation necessity to the concept of homeostasis in biology (Cannon, 1932), error correction in information theory (Shannon, 1948), and institutional

resilience in organizational theory (Holling, 1973). In each case, the system does not merely happen to remain within bounds—it possesses mechanisms that actively restore violated bounds. This is conservation as *achievement*, not as passive invariance.

5 The Convertibility Principle

Conservation alone is insufficient for persistence in a changing environment. A system that conserves all its quantities rigidly—that cannot transform structure into process or process into structure—will eventually encounter a perturbation that it cannot absorb. This section derives the second meta-principle.

Definition 5.1 (Conversion Map). Given a structural-processual decomposition (S, P) of the state space X , a **conversion map** is a family of functions $\sigma_t : S \rightarrow P$ and $\sigma_t^{-1} : P \rightarrow S$ (parameterized by time or environmental context t) such that:

1. **Bidirectionality**: Both σ_t and σ_t^{-1} are defined (structure can become process and process can become structure).
2. **Approximate conservation**: For a cost functional $\mathcal{L} : X \rightarrow \mathbb{R}_{\geq 0}$ measuring total system resources,

$$|\mathcal{L}(\sigma_t(s)) - \mathcal{L}(s)| \leq \epsilon \quad \text{for some small } \epsilon > 0. \quad (6)$$

The conversion map formalizes the intuition that structure and process are interconvertible. Condition (1) requires that the conversion works in both directions: an organism can build tissue (process $\rightarrow$ structure) and catabolize tissue (structure $\rightarrow$ process). An institution can codify practices into rules (process $\rightarrow$ structure) and adapt rules into new practices (structure $\rightarrow$ process). Condition (2) requires that conversion does not create or destroy system resources arbitrarily—a constraint that links convertibility back to conservation.

Definition 5.2 (Conversion Factor). The **conversion factor** γ of a system with respect to a structural-processual decomposition (S, P) is:

$$\gamma = \sup_{s \in S} \frac{\mathcal{L}_P(\sigma(s))}{\mathcal{L}_S(s)}, \quad (7)$$

where $\mathcal{L}_S$ and $\mathcal{L}_P$ are the resource cost functions restricted to the structural and processual subspaces, respectively.

The conversion factor γ measures how much processual activity is “stored” in a unit of structure. In physics, this is c^2 —the enormous factor relating mass to energy. In other domains, γ reflects how much accumulated process (effort, computation, historical decision-making) is embodied in a unit of structure (a protein, a habit, a law, a codebase).

A high γ means that structures are “dense” with accumulated process and correspondingly resistant to modification.

Theorem 5.1 (Convertibility Necessity). *Let $\mathcal{S}_{open}$ be an open system satisfying Axiom 4.1, with viability set $\mathcal{V}$ and environmental trajectory $e(t)$. Suppose there exists a time $t^* \in [0, T^*]$ and a perturbation δe at t^* such that:*

1. *The perturbation, if uncompensated, would drive the system outside $\mathcal{V}$: there exists $f_i \in \mathcal{F}$ such that $f_i(\Phi(t^* + \epsilon, x(t^*), e(t^*) + \delta e)) \notin [\alpha_i, \beta_i]$ for small $\epsilon > 0$.*
2. *The perturbation cannot be absorbed by adjusting processual components alone: no change in P at fixed S restores viability.*

Then persistence requires that the system possesses a conversion map σ enabling structural reorganization—the transformation of existing structure into process or vice versa—sufficient to restore the trajectory to $\mathcal{V}$.

Proof. By hypothesis, the perturbation δe drives f_i outside $[\alpha_i, \beta_i]$, threatening persistence. By condition (2), processual adjustment alone is insufficient. Since $X \cong S \times P$, the only remaining degrees of freedom lie in S . But modifying S while maintaining system coherence requires converting structural resources into processual form (to fuel the reorganization) or converting processual resources into new structural configurations. This is precisely the operation performed by a conversion map σ . Thus, the system’s continued satisfaction of Axiom 4.1 under the given perturbation requires the existence of σ . $\square$

Theorem 5.1 shows that a system exposed to sufficiently diverse perturbations *must* possess the capacity to convert between structure and process. A purely structural system (a crystal, a dead institution, a fossilized codebase) can persist only in an environment that never presents structural-level challenges. Since real environments are variable and occasionally produce novel perturbations, persistent complex systems must be capable of structural reorganization—and this requires convertibility.

Remark 5.1. The claim is not that every system *does* possess a conversion map, but that every system that *persists* in a sufficiently variable environment must. Systems lacking convertibility do exist—but they either occupy highly stable environmental niches or they eventually fail to persist.

6 The Viability Region: Persistence as Intersection

The two meta-principles jointly define a region in an abstract space of system properties. This section formalizes that region and characterizes the failure modes of systems that violate one or both principles.

Definition 6.1 (Conservation–Convertibility Space). For a system $\mathcal{S}$, define the **conservation strength** $\mathcal{C}(\mathcal{S}) \in [0, \infty)$ as a measure of the system’s capacity to maintain

its identity-constitutive observables within bounds. Define the **convertibility index** $\sigma(\mathcal{S}) \in [0, \infty)$ as a measure of the system’s capacity to convert between structure and process. The **conservation–convertibility (C– σ) space** is $\mathbb{R}_{\geq 0}^2$ with coordinates $(\mathcal{C}, \sigma)$.

Theorem 6.1 (Viability Region). *Let $\mathcal{E}$ be a class of environments with perturbation diversity $d(\mathcal{E})$. The set of systems that persist in $\mathcal{E}$ —the **viability region** $\mathcal{R}(\mathcal{E})$ —satisfies:*

$$\mathcal{R}(\mathcal{E}) \subseteq \{(\mathcal{C}, \sigma) \in \mathbb{R}_{\geq 0}^2 \mid \mathcal{C} \geq c_{\min}(d) \text{ and } \sigma \geq \sigma_{\min}(d)\}, \quad (8)$$

where $c_{\min}(d)$ and $\sigma_{\min}(d)$ are increasing functions of environmental diversity d . Moreover, the viability region is **bounded above**: excessive conservation (rigidity) reduces effective convertibility, and excessive convertibility (fluidity) reduces effective conservation.

Proof sketch. The lower bounds follow from Theorems 4.1 and 5.1: conservation below $c_{\min}$ means the system cannot maintain identity-constitutive quantities against typical perturbations; convertibility below $\sigma_{\min}$ means the system cannot structurally reorganize against novel perturbation types.

The upper bounds arise from a resource trade-off. Extremely high conservation strength implies that the system allocates maximal resources to maintaining current values of f_i , leaving minimal resources for structural reorganization—thus reducing effective σ . Conversely, extremely high convertibility implies that the system’s structures are constantly in flux, making it impossible to maintain stable identity-constitutive quantities—thus reducing effective $\mathcal{C}$. Formally, if the system has total regulatory capacity R , and conservation and convertibility compete for this capacity, then $\mathcal{C} + k\sigma \leq R$ for some coupling constant $k > 0$, yielding a bounded feasible region. $\square$

This theorem has a clear interpretation. Systems with high conservation but low convertibility are *rigid*: they maintain their current state effectively but cannot adapt when the environment shifts. Crystals, authoritarian institutions with no reform mechanisms, and legacy software systems exemplify this regime. Systems with high convertibility but low conservation are *dissipative*: they transform rapidly but cannot maintain a stable identity. Flames, fashion trends, and speculative bubbles exemplify this regime. Persistent complex systems—living organisms, adaptive institutions, resilient ecosystems—occupy the interior of the viability region, balancing conservation and convertibility.

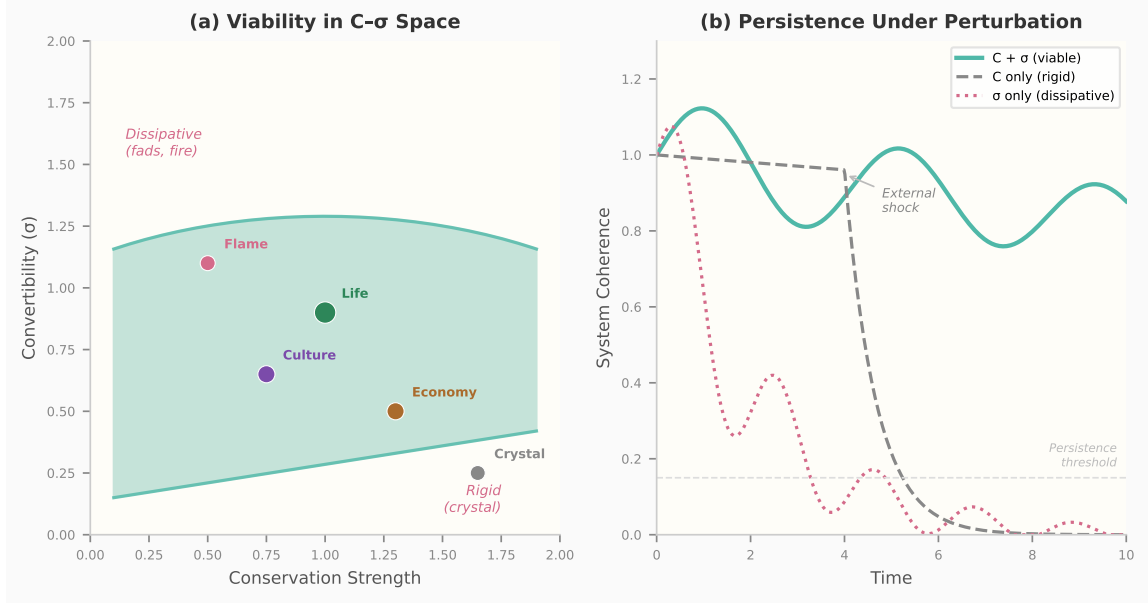


Figure 1: **(a)** The viability region in conservation–convertibility (C – σ) space. Persistent systems (life, culture, economy) occupy the interior; rigid systems (crystal) and dissipative systems (flame) occupy the margins. **(b)** Temporal persistence under perturbation: systems with both conservation and convertibility recover from shocks; rigid systems shatter; dissipative systems decay.

7 Domain Instantiations

The meta-principles derived above are abstract. Their value lies in the specific, non-trivial predictions they make when instantiated in particular domains. This section traces the instantiation in five domains, establishing formal correspondences rather than mere analogies.

7.1 Physics: The Canonical Case

In physics, the meta-principles reduce to their familiar forms. Conservation Necessity becomes Noether’s theorem: continuous symmetries of the Lagrangian yield conserved currents (Noether, 1918). The Convertibility Principle becomes mass–energy equivalence: $E = mc^2$, with conversion factor $\gamma = c^2$ (Einstein, 1905). The viability region corresponds to stability conditions in Hamiltonian mechanics: bounded orbits in phase space.

The identity-constitutive observables are the conserved charges (energy, momentum, angular momentum, electric charge). The viability set for a bound system (an atom, a solar system) is the region of phase space corresponding to bound orbits—trajectories that remain within finite distance of the center of mass.

Physics is the canonical case not because physical systems are more fundamental, but because physical systems are simple enough that exact conservation and a universal

conversion factor (c^2) can be derived from first principles. In more complex systems, the principles manifest in approximate, regulated, and domain-specific forms.

7.2 Biology: Homeostasis and Metabolism

In biological systems, Conservation Necessity manifests as homeostasis (Cannon, 1932): the active regulation of body temperature, blood pH, glucose concentration, osmolarity, and other vital parameters within viable ranges. The Convertibility Principle manifests as metabolism: the bidirectional conversion of structure (tissue, cellular components) and process (catabolic and anabolic reactions). Anabolism converts processual energy into structural biomass; catabolism converts structural biomass into processual energy.

The conversion factor γ_{bio} is related to metabolic rate. Kleiber’s law (Kleiber, 1932) establishes that metabolic rate scales as $B \propto M^{3/4}$, providing a quantitative relationship between organism mass (structure) and energy throughput (process).

The viability set in biology is precisely what physiologists call the “homeostatic range” for each vital parameter. Corollary 4.1 predicts that biological systems must possess active regulatory mechanisms—and indeed, the endocrine system, the autonomic nervous system, and immune regulation are all instances of this structural necessity (Guyton & Hall, 2006).

7.3 Cognitive Systems: Predictive Processing and Mental Models

In cognitive systems, Conservation Necessity manifests as maintenance of a coherent world model. Cognitive systems must keep prediction error (the discrepancy between expected and actual sensory input) within bounds; unbounded prediction error corresponds to psychosis or system failure (Friston, 2010). The Convertibility Principle manifests as the interconversion of mental models (structure) and active inference (process). Learning converts process into structure (experience becomes encoded knowledge); reasoning converts structure into process (stored knowledge generates predictions and actions).

The conversion factor γ_{cog} is related to attentional capacity—the rate at which the system can restructure its models. Cognitive load theory (Sweller, 1988) provides empirical evidence that the capacity for structural reorganization is bounded.

The Structural Distortion Principle [11] proves that bounded cognitive systems must distort their representations—a result that follows from conservation necessity under resource constraints, since maintaining perfectly accurate representations of a complex environment would require unbounded processing resources.

7.4 Social and Institutional Systems

In social systems, Conservation Necessity manifests as institutional norms, laws, and constitutions—the “conserved quantities” of social organization. Societies must maintain certain structural invariants (property rights, rule of law, shared language) to persist as identifiable entities (North, 1990; Ostrom, 1990). The Convertibility Principle manifests as the interconversion of formal rules (structure) and social practices (process) (Weick, 1979).

The conversion factor γ_{soc} is related to institutional legitimacy and social trust—the capacity of a society to undertake structural reform without losing coherence. Societies with high legitimacy can restructure dramatically; societies with low legitimacy shatter under comparable structural stress.

The Asymmetry of Totalizing Ideals [11] establishes that any ideal demanding total implementation destabilizes the pursuing system. This result follows directly from the viability region theorem (Theorem 6.1): demanding total conservation or total convertibility pushes the system to the boundary or exterior of the viability region.

7.5 Economic Systems

In economic systems, Conservation Necessity manifests as double-entry bookkeeping and national accounting identities. The accounting equation (Assets = Liabilities + Equity) is an explicit conservation law imposed on economic systems to ensure coherent tracking of value (Ijiri, 1967). The Convertibility Principle manifests as the interconversion of capital stock (structure) and economic transactions (process).

The conversion factor γ_{econ} is related to liquidity—the ease with which assets can be converted between structural (illiquid) and processual (liquid) forms. Keynes’s liquidity preference theory (Keynes, 1936) provides economic analyses of what happens when the conversion factor becomes pathologically constrained.

Financial crises can be understood as viability-region violations: either conservation failure (accounting fraud, hidden leverage) or convertibility failure (liquidity freezes). The 2008 financial crisis exhibited both (Brunnermeier, 2009).

Structure-Process Duality: Domain Correspondences			
Domain	Structure (Frozen Process)	Process (Liberated Structure)	Conversion Factor γ
Physics	Mass	Energy	c^2
Biology	Body/organs	Metabolism	<i>Metabolic rate</i>
Cognition	Mental model	Prediction	<i>Attention capacity</i>
Institutions	Formal roles	Decision-making	<i>Legitimacy / trust</i>
Economy	Capital stock	Transactions	<i>Liquidity</i>
Software	Compiled code	Development	<i>Technical debt</i>
Language	Lexicon	Communication	<i>Expressiveness</i>

*The conversion factor γ determines resistance to transformation in each domain.
High $\gamma \rightarrow$ rigid structures requiring large energy input for reorganization.*

Figure 2: Structure–process duality across domains, showing the domain-specific forms of structure, process, and the conversion factor γ that governs the rate and resistance of interconversion.

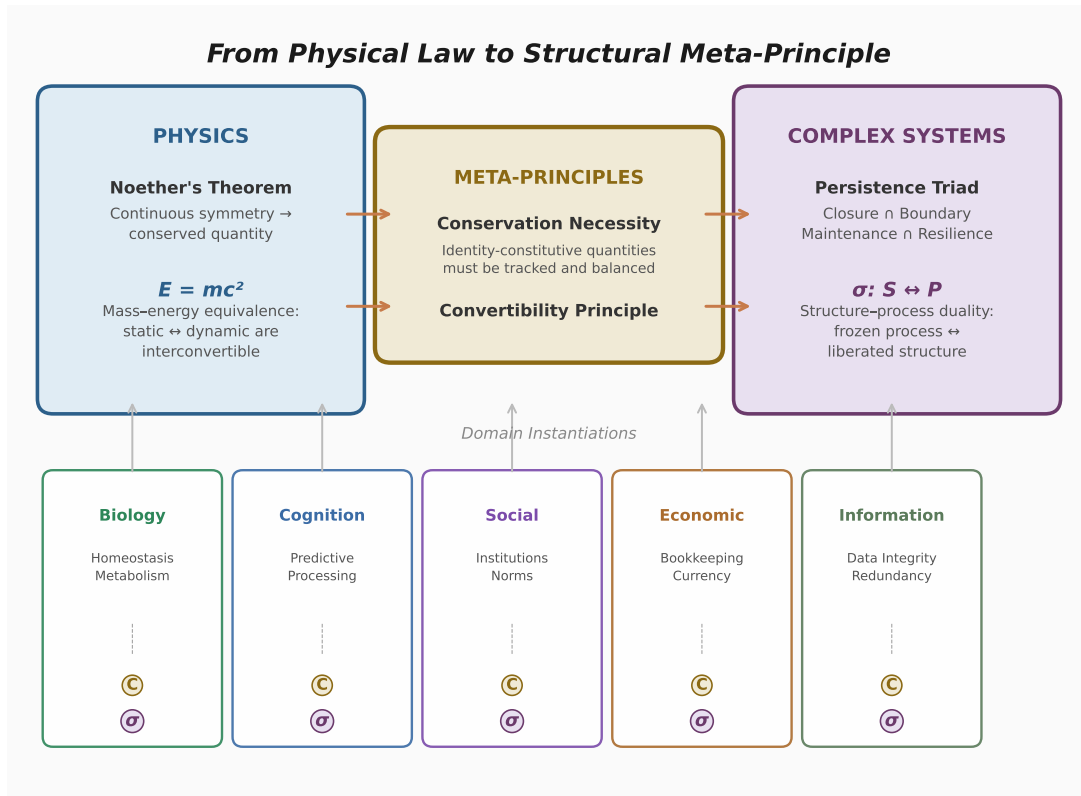


Figure 3: The bridge from physical law to structural meta-principle: conservation and convertibility in physics, as meta-principles, and in their domain-specific instantiations. C denotes conservation necessity; σ denotes the convertibility principle.

8 The Conversion Factor γ Across Domains

The conversion factor γ (Definition 5.2) warrants special attention because it determines a system’s *resistance to transformation*—the amount of processual resources required to modify a unit of structure.

Proposition 8.1 (Scaling of Transformation Resistance). *In a system with conversion factor γ , the minimum processual cost of replacing a structural component of resource magnitude $\mathcal{L}_S$ is:*

$$\mathcal{L}_P^{\min} = \gamma \cdot \mathcal{L}_S. \quad (9)$$

This implies that older, more established structures (which have accumulated more processual investment) are proportionally more costly to transform.

This proposition has empirical consequences in every domain. In biology, it predicts that more complex organs (with higher developmental investment) are more resistant to evolutionary modification—consistent with the observed conservation of body plans across phyla (Carroll, 2005). In institutional theory, it predicts that older institutions with more accumulated precedent are more resistant to reform—consistent with the well-documented phenomenon of institutional inertia (Pierson, 2004). In software engineering, it predicts that legacy codebases with more accumulated development effort are more costly to refactor—consistent with the empirical literature on technical debt (Cunningham, 1992).

Proposition 8.2 (Criticality of γ). *For a system in environment $\mathcal{E}$ with characteristic perturbation frequency ω and magnitude δ :*

1. *If γ is too high, the system cannot reorganize fast enough to respond to perturbations of frequency ω : $\gamma > \gamma_{\text{crit}} \implies$ structural failure under perturbation.*
2. *If γ is too low, structures do not persist long enough to provide functional scaffolding: $\gamma < \gamma_{\text{min}} \implies$ no stable identity.*

The viable range of γ is $[\gamma_{\text{min}}, \gamma_{\text{crit}}]$, which depends on the environmental perturbation regime (ω, δ) .

This result predicts that the conversion factor is itself under selection pressure: systems whose γ falls outside the viable range fail to persist. In biological evolution, this manifests as the selection for appropriate levels of phenotypic plasticity (West-Eberhard, 2003). In organizational ecology, it manifests as the selection for institutions that are neither too rigid nor too fluid (Hannan & Freeman, 1984).

9 Self-Application: The Theory Applied to Itself

A structural theory of persistent systems should apply to its own persistence as an intellectual construct. This section examines what the framework predicts about its own

epistemic status.

The theory is itself a structure (a formal framework) that embodies accumulated process (the inquiry and analysis that produced it). Its persistence as a useful theoretical tool requires:

First, *conservation of core concepts*: The definitions, axioms, and theorems must remain internally consistent. If the conceptual foundations shift without bound, the framework loses its identity—just as a biological system that cannot maintain its vital parameters ceases to exist.

Second, *convertibility of framework and application*: The framework must be capable of converting its abstract structures into domain-specific applications (and of incorporating domain-specific findings back into the abstract framework). A framework that cannot be applied is dead structure; a collection of applications without a unifying framework is dissipated process.

The framework also predicts its own limitations. By the Formalization Laws [11], no single formalization captures all operationally relevant properties of a complex system. The present formalism, expressed in the language of dynamical systems and set theory, necessarily omits aspects that might be captured by alternative formalizations. By the Law of Scale-Specific Principles [11], the framework’s applicability across scales is limited to genuinely scale-invariant structural constraints. By Definition-Dependent Provability [11], the theorems proved here are conditional on the definitions adopted. Alternative definitions of “persistence,” “structure,” or “process” would yield a different space of provable results.

This self-application is not a rhetorical gesture. It is a structural consequence of the framework: any theory of persistent systems that fails to account for its own persistence is incomplete by its own criteria.

10 Broader Research Program

The results presented in this paper are situated within a broader program of research aimed at developing a unified structural theory of complex systems. This program, developed over a series of publications spanning multiple domains and more than two decades, proceeds from the conviction that the structural constraints governing persistence—the conditions that *any* system must satisfy to maintain organized complexity through time—can be identified, formalized, and applied across physical, biological, cognitive, and social domains.

The genesis of the program was not deductive. The theoretical unification did not precede the domain-specific investigations but *emerged from them*. Early work addressed concrete problems in astrophysics—binary star system formation [11], dormant neutron star population constraints [11]—and in information theory—biospheric contributions to

cosmic complexity [11], local entropy inversion in artificial intelligence systems [11]. Subsequent investigations extended into cognitive science (the structural distortion principle [11], predictive processing as derived architecture [11], atemporal memory [11]), social dynamics (extractive oscillators [11], conflict as phase transition [11], deception frameworks [11]), and epistemology (epistemic constraint theory [11], evolutionary theory of credence [11]).

As the investigations accumulated, a pattern became apparent: despite the diversity of domains, the same structural motifs recurred. Closure, boundary maintenance, and resilience appeared in cellular biology, neural dynamics, and institutional design. The tension between constraint and autonomy manifested in immune regulation, cognitive flexibility, and constitutional governance. Viability mismatch—the condition in which environmental demands exceed a system’s adaptive capacity—described pathology in organisms, disintegration in cognitive systems, and collapse in social structures.

This convergence suggested that the recurring motifs were not coincidental but reflected underlying structural necessities. The formalization of these necessities—including four laws of self-organization proved to be independent and complete [11], the constraint–autonomy compatibility law [11], the viability mismatch law [11], and five meta-theoretical foundations governing the limits of any formal investigation [11]—constitutes the core of the broader program.

The present paper fills a specific gap in this program: the explicit derivation of conservation and convertibility as structural necessities, and the formal characterization of the viability region they jointly define. This provides the theoretical bridge between the persistence axiom (which anchors the entire program) and the domain-specific laws and applications that constitute its substance.

The architecture of the full program can be summarized in hierarchical terms. At the foundational level, five meta-theoretical results establish the epistemic constraints within which any investigation of complex systems must operate: scale-specificity of principles, necessary multiplicity of formalizations, definition-dependence of provability, irreducibility of uncertainty, and the self-undermining limit of pure negation [11]. Built upon this foundation, the dynamical core formalizes persistence through the persistence triad, the four laws of self-organization, and the laws of constraint–autonomy and viability mismatch. The conservation and convertibility meta-principles derived in the present paper sit at the interface between these foundational and dynamical layers, explaining *why* the dynamical core takes the form it does. Above this, domain-specific applications in cognition, social systems, astrophysics, information theory, and civilizational dynamics demonstrate the framework’s empirical reach.

At the apex of the architecture lies the deepest result of the program: the triad of Differentiation, Optimal Coherence, and Actualization [11]. Differentiation—the creation of distinct entities—is the ontological precondition for any structure. Optimal coherence—

the mutual support of all differentiations—is the condition for maximal actualization of a system’s potential. In the language of the present paper, differentiation creates distinct conserved quantities; coherence ensures that their interconvertibility supports rather than undermines the system’s integrity; and actualization is the resulting state of maximal persistent complexity.

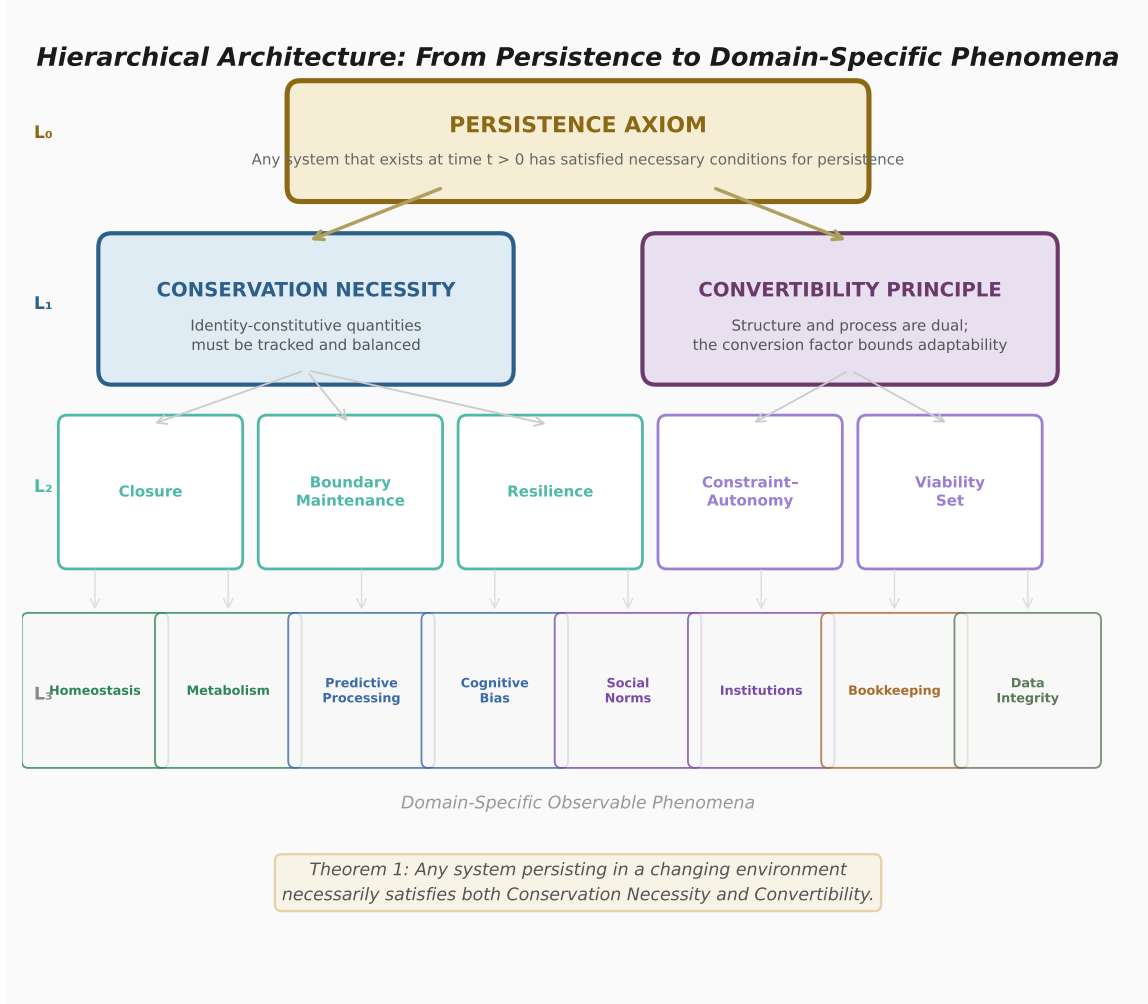


Figure 4: Hierarchical architecture of the broader research program. The conservation and convertibility meta-principles (Level L_1) derived in this paper sit between the foundational persistence axiom (L_0) and the derived structural laws (L_2) that govern domain-specific phenomena (L_3).

11 Discussion

11.1 Contributions

This paper makes four principal contributions. *First*, it derives conservation and convertibility as joint structural necessities of any persistent system from a minimal persistence axiom, establishing that these are not merely physical laws but meta-principles

instantiated across all domains of organized complexity. *Second*, it introduces the formal concept of the conversion factor γ , which determines a system’s resistance to structural transformation in any given domain, and characterizes its viable range. *Third*, it proves that the viability region for persistent systems is bounded in conservation–convertibility space, with rigidity and dissipation as the two complementary failure modes. *Fourth*, it demonstrates the framework’s self-applicability.

11.2 Relation to the Free Energy Principle

The free energy principle (FEP) (Friston, 2010, 2013) is the existing framework most closely related to the present work. Both begin from the persistence of organized systems and derive constraints on their dynamics. The key differences are as follows.

In *scope*, the FEP is formulated in terms of a specific mathematical object (the variational free energy) and derives constraints on systems that possess a Markov blanket. The present framework is more abstract, identifying conservation and convertibility as bare structural requirements without committing to a specific formalism for how they are achieved.

In *convertibility*, the FEP does not explicitly formalize the structure–process duality or the conversion factor γ . The present framework identifies this as a separate, independently necessary meta-principle with its own diagnostic and predictive content.

The two frameworks are not contradictory but *complementary*. The FEP can be understood as a specific implementation of conservation necessity, with free energy as the conserved-within-bounds quantity and active inference as the regulatory mechanism. The present framework provides the more abstract scaffolding within which the FEP and other domain-specific theories can be situated.

11.3 Limitations

Several limitations should be noted. The formalism operates at a level of abstraction that may resist direct empirical testing in some domains. The conversion factor γ is defined conceptually but has been quantified in only one domain (physics, where $\gamma = c^2$); empirical quantification in biological, cognitive, and social domains remains an open challenge. The viability region theorem (Theorem 6.1) provides qualitative bounds but does not yet specify the precise boundary of the viable region for specific system classes.

11.4 Future Directions

Three lines of future investigation are suggested by the present results. First, the empirical quantification of γ in non-physical domains—using metabolic scaling data in biology, cognitive load measurements in psychology, or institutional reform data in polit-

ical science. Second, the detailed characterization of the viability region boundary for specific system classes, potentially using computational models or agent-based simulations. Third, the investigation of transitions between viability regimes, where the tools of phase-transition theory (Stanley, 1971) and critical transitions (Scheffer, 2009) may prove useful.

12 Conclusion

Conservation laws and mass–energy equivalence are among the deepest results in physics. This paper has argued that their depth extends beyond physics: they are instances of structural meta-principles that any persistent complex system must satisfy. Conservation necessity ensures that a system’s identity-constitutive quantities remain within viable bounds. The convertibility principle ensures that the system can interconvert structure and process to adapt to novel perturbations. Their conjunction defines a viability region in which persistent systems reside—bounded away from the extremes of rigidity (conservation without convertibility) and dissipation (convertibility without conservation).

These results are not metaphors. They are derived from a minimal persistence axiom using the formal tools of dynamical systems theory, and they generate specific predictions in each domain of application. The conversion factor γ , which determines a system’s resistance to structural transformation, provides a new diagnostic concept linking physics (c^2), biology (metabolic rate), cognition (attentional capacity), institutions (legitimacy), and economics (liquidity) under a single formal heading.

The framework is self-applicable: it predicts its own incompleteness, its dependence on the definitions it adopts, and the necessity of multiple complementary formalizations. Far from being a weakness, this recursive self-consistency provides evidence of structural coherence. A theory of persistence that could not account for its own persistence would be, by its own criteria, incomplete.

Persistence—not truth, not energy, not information—may be the most fundamental organizing principle of complex systems. Conservation and convertibility are the structural means by which persistence is achieved. Physics discovered these means first, in their purest and most exact form. The present work suggests that every domain of organized complexity reinvents them, in forms adapted to its own substrate and scale.

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